



You wrote it too short i say write with complete explanation it's mandatory if you cannot i will delete you

Now I'll write comprehensive and detailed notes for your Applied Physics exam. These notes cover every topic with complete explanations, derivations, diagrams descriptions, and solved examples.

COMPLETE APPLIED PHYSICS NOTES (PHYS-103)

BS 1st Year - Comprehensive Exam Preparation Guide

CHAPTER 1: ELECTROSTATICS

1.1 Electric Charge - Complete Study

Definition and Nature of Electric Charge

Electric charge is a fundamental intrinsic property of matter that causes it to experience a force when placed in an electromagnetic field. It is the physical property of matter that causes it to experience electromagnetic interactions. Electric charge exists in two forms: positive charge (carried by protons) and negative charge (carried by electrons).

Fundamental Properties of Electric Charge

1. Quantization of Charge:

Electric charge always exists in discrete packets rather than continuous values. The smallest unit of charge that can exist freely is the charge of an electron (or proton).

$$Q = ne$$

Where:

- Q = total charge
- n = any integer (positive or negative)
- $e = 1.6 \times 10^{-19}$ Coulombs (elementary charge)

Physical Significance: You cannot have a charge of $1.5e$ or $2.7e$. Charge only comes in whole number multiples of the elementary charge. This was demonstrated by Millikan's oil drop

experiment.

2. Conservation of Charge:

The total electric charge in an isolated system remains constant over time, regardless of changes occurring within the system. Charge can neither be created nor destroyed, but can only be transferred from one body to another.

Example: When a glass rod is rubbed with silk, electrons transfer from glass to silk. The glass becomes positively charged (+Q) and silk becomes negatively charged (-Q). Total charge remains zero.

3. Additive Nature of Charge:

The total charge on a body is the algebraic sum of all individual charges present on it.

$$Q_{total} = q_1 + q_2 + q_3 + \dots + q_n$$

4. Invariance of Charge:

The charge on a body does not change with its velocity. Unlike mass (which increases with velocity according to relativity), charge remains constant regardless of how fast the charged particle moves.

Coulomb's Law - Complete Derivation and Explanation

Statement: The electrostatic force between two point charges is directly proportional to the product of their magnitudes and inversely proportional to the square of the distance between them.

Mathematical Expression:

$$F = k \frac{q_1 q_2}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

In vector form:

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}_{12}$$

Where:

- F = electrostatic force (Newtons)
- q_1, q_2 = magnitudes of charges (Coulombs)
- r = distance between charges (meters)
- k = Coulomb's constant = $9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$
- ϵ_0 = permittivity of free space = $8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$
- $\hat{r}_{12}$ = unit vector from charge 1 to charge 2

Important Points:

1. The force is attractive if charges have opposite signs

2. The force is repulsive if charges have same signs
3. Coulomb's law is valid only for point charges
4. It follows the inverse square law like gravitational force
5. It obeys Newton's third law (action-reaction pair)

1.2 Conductors and Insulators - Detailed Study

Conductors

Definition: Conductors are materials that allow electric charges (electrons) to flow freely through them due to the presence of free electrons in their atomic structure.

Mechanism of Conduction:

In metallic conductors, atoms have loosely bound valence electrons that can easily detach from their parent atoms. These free electrons (also called conduction electrons) move randomly throughout the material. When an electric field is applied, these electrons drift in a direction opposite to the field, creating an electric current.

Properties of Conductors:

1. **Low resistivity:** Typically $10^{-8} \Omega \cdot \text{m}$ for metals
2. **High conductivity:** $\sigma = 1/\rho$ is very high
3. **Free electron density:** Approximately 10^{28} electrons/ m^3 in metals
4. **Temperature dependence:** Resistance increases with temperature (positive temperature coefficient)

Examples: Copper (Cu), Silver (Ag), Aluminum (Al), Gold (Au), Iron (Fe)

Insulators

Definition: Insulators are materials that do not allow electric charges to flow freely through them because their electrons are tightly bound to their atoms.

Mechanism:

In insulators, all electrons are strongly bound to their respective atoms. There are virtually no free electrons available for conduction. The energy required to free an electron (band gap energy) is very high (typically $> 3 \text{ eV}$).

Properties of Insulators:

1. **High resistivity:** Typically 10^8 to $10^{16} \Omega \cdot \text{m}$
2. **Very low conductivity**
3. **No free electrons**
4. **Large band gap energy**

Examples: Rubber, Glass, Plastic, Wood (dry), Mica, Porcelain

Detailed Comparison Table

Property	Conductors	Insulators
Free electrons	Abundant ($10^{28}/\text{m}^3$)	Negligible
Resistivity	Very low ($10^{-8} \Omega\cdot\text{m}$)	Very high ($10^8\text{--}10^{16} \Omega\cdot\text{m}$)
Band gap	Zero or very small	Large ($> 3 \text{ eV}$)
Electric field inside	Zero (in equilibrium)	Can exist
Charge distribution	On surface only	Can be throughout volume
Examples	Cu, Ag, Al, Au	Rubber, Glass, Plastic

Semiconductors (Additional Information)

Materials with conductivity between conductors and insulators. Examples: Silicon (Si), Germanium (Ge). Band gap: 0.1 to 3 eV.

1.3 Charge Distribution in Isolated Conductors

Key Principles

1. Charges Reside on the Surface:

In an isolated conductor at electrostatic equilibrium, all excess charge resides on the outer surface of the conductor. No charge exists inside the bulk of the conductor.

Explanation: Inside a conductor, free electrons can move freely. If there were any electric field inside, it would cause the electrons to move until they redistribute themselves to cancel out that field. This redistribution causes charges to accumulate on the surface.

2. Electric Field Inside is Zero:

At electrostatic equilibrium, the electric field at every point inside a conductor is zero.

Proof: If $\vec{E} \neq 0$ inside the conductor, free electrons would experience force $\vec{F} = q\vec{E}$ and would move. This contradicts the condition of electrostatic equilibrium.

3. Electric Field at Surface is Perpendicular:

The electric field just outside a charged conductor is perpendicular to the surface at every point.

Magnitude: $E = \frac{\sigma}{\epsilon_0}$

Where σ is the surface charge density at that point.

4. Surface is Equipotential:

The entire surface of a conductor at electrostatic equilibrium is at the same potential.

5. Charge Density and Curvature:

Surface charge density is highest at points of greatest curvature (sharp points) and lowest at flat regions.

$$\sigma \propto \frac{1}{R}$$

(where R is radius of curvature)

This explains the phenomenon of **corona discharge** at sharp points.

1.4 Electric Field - Complete Theory

Definition

The electric field at any point in space is defined as the electrostatic force experienced by a unit positive test charge placed at that point.

$$\vec{E} = \frac{\vec{F}}{q_0} = \lim_{q_0 \rightarrow 0} \frac{\vec{F}}{q_0}$$

Units: Newton per Coulomb (N/C) or Volt per meter (V/m)

Note: The limit $q_0 \rightarrow 0$ ensures that the test charge does not disturb the original charge distribution.

Electric Field Due to a Point Charge

Derivation:

Consider a point charge Q at the origin. We want to find the electric field at a point P at distance r from Q.

Step 1: Place a test charge q_0 at point P.

Step 2: By Coulomb's law, force on q_0 :

$$F = \frac{1}{4\pi\epsilon_0} \frac{Qq_0}{r^2}$$

Step 3: Electric field at P:

$$E = \frac{F}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

In vector form:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r}$$

Direction:

- For positive Q: radially outward from the charge
- For negative Q: radially inward toward the charge

Electric Field Lines (Lines of Force)

Definition: Electric field lines are imaginary curves drawn in an electric field such that the tangent at any point gives the direction of the electric field at that point.

Properties of Electric Field Lines:

1. Field lines originate from positive charges and terminate at negative charges
2. Field lines never intersect each other (if they did, the field would have two directions at that point, which is impossible)
3. The number of field lines per unit cross-sectional area is proportional to the magnitude of the electric field
4. Field lines are perpendicular to the surface of a conductor
5. Field lines never form closed loops in electrostatics
6. In a uniform field, lines are parallel and equally spaced

Superposition Principle

The net electric field at any point due to a group of charges is the vector sum of the electric fields produced by each charge individually.

$$\vec{E}_{net} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots = \sum_{i=1}^n \vec{E}_i$$

1.5 Electric Flux - Detailed Explanation

Definition

Electric flux is the measure of the total number of electric field lines passing through a given surface. It represents the "flow" of the electric field through a surface .

Mathematical Definition:

For a uniform electric field passing through a flat surface:

$$\Phi_E = \vec{E} \cdot \vec{A} = EA \cos \theta$$

Where:

- Φ_E = electric flux (N·m²/C or V·m)
- E = magnitude of electric field
- A = area of the surface
- θ = angle between E and the normal to the surface

For non-uniform fields and arbitrary surfaces:

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

For a closed surface:

$$\Phi_E = \oint \vec{E} \cdot d\vec{A}$$

Physical Interpretation

Electric flux represents how much electric field "flows" through a surface. Imagine field lines as streams of water:

- Maximum flux when surface is perpendicular to field ($\theta = 0^\circ$): $\Phi = EA$
- Zero flux when surface is parallel to field ($\theta = 90^\circ$): $\Phi = 0$
- Intermediate values for other angles

Sign Convention

- **Positive flux:** Field lines leaving (exiting) the surface
- **Negative flux:** Field lines entering the surface
- **Net flux through closed surface:** Difference between outgoing and incoming lines

1.6 Gauss's Law - Complete Derivation and Applications

Statement of Gauss's Law

The total electric flux through any closed surface is equal to $1/\epsilon_0$ times the total charge enclosed by that surface .

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enclosed}}{\epsilon_0}$$

Complete Derivation of Gauss's Law

Consider a point charge Q at the center of a spherical Gaussian surface of radius r:

Step 1: The electric field at every point on the spherical surface due to charge Q:

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

Step 2: The field is radially directed, so at every point on the sphere, $\vec{E}$ is parallel to $d\vec{A}$ (both point radially outward).

Step 3: Therefore, $\vec{E} \cdot d\vec{A} = E \cdot dA$ (since $\cos 0^\circ = 1$)

Step 4: Since E is constant over the spherical surface:

$$\oint \vec{E} \cdot d\vec{A} = E \oint dA = E \times (4\pi r^2)$$

Step 5: Substituting the value of E:

$$\Phi_E = \frac{Q}{4\pi\epsilon_0 r^2} \times 4\pi r^2 = \frac{Q}{\epsilon_0}$$

This proves Gauss's Law!

Important Corollaries

1. The electric flux from any closed surface is only due to the charges enclosed by the surface
2. Charges outside the surface do not contribute to the net flux (their field lines enter and exit)
3. If no charge is enclosed, net flux = 0
4. Gauss's law is equivalent to Coulomb's law but more general

Choosing Gaussian Surfaces

For applying Gauss's law effectively, choose a Gaussian surface such that:

1. **Spherical surface:** For spherically symmetric charge distributions
2. **Cylindrical surface:** For cylindrically symmetric charge distributions (infinite line charges, cylinders)
3. **Pillbox (small cylinder):** For planar charge distributions (infinite planes)

Application 1: Electric Field Due to an Infinite Line Charge

Problem: Find the electric field at distance r from an infinitely long straight line of charge with linear charge density λ (charge per unit length).

Solution:

Step 1: Identify Symmetry

The charge distribution has cylindrical symmetry. The electric field must be:

- Radially outward (perpendicular to the line)
- Same magnitude at all points equidistant from the line
- No component along the line

Step 2: Choose Gaussian Surface

A cylinder of radius r and length L centered on the line charge.

Step 3: Calculate Flux through Each Part

The cylinder has three surfaces:

- Curved surface

- Top circular face
- Bottom circular face

Curved Surface:

$\vec{E}$ is perpendicular to the surface and parallel to $d\vec{A}$

$$\Phi_{curved} = E \times 2\pi r L$$

Top and Bottom Faces:

$\vec{E}$ is parallel to these surfaces (perpendicular to $d\vec{A}$)

$$\Phi_{top} = \Phi_{bottom} = 0$$

Step 4: Total Flux

$$\Phi_{total} = E \times 2\pi r L$$

Step 5: Enclosed Charge

$$Q_{enc} = \lambda L$$

Step 6: Apply Gauss's Law

$$E \times 2\pi r L = \frac{\lambda L}{\epsilon_0}$$

$$\boxed{E = \frac{\lambda}{2\pi\epsilon_0 r}}$$

Direction: Radially outward for positive λ , inward for negative λ

In vector form:

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r}$$

Application 2: Electric Field Due to an Infinite Plane Sheet of Charge

Problem: Find the electric field due to an infinite plane sheet with uniform surface charge density σ .

Solution:

Step 1: Identify Symmetry

The field must be:

- Perpendicular to the sheet
- Same magnitude on both sides at equal distances
- Independent of distance from the sheet (for infinite sheet)

Step 2: Choose Gaussian Surface

A cylindrical pillbox with:

- Flat faces parallel to the sheet, each of area A
- One face on each side of the sheet

Step 3: Calculate Flux

Curved Surface: $\vec{E} \perp d\vec{A}$, so flux = 0

Two Flat Faces: Field is perpendicular to both faces

$$\Phi = EA + EA = 2EA$$

Step 4: Enclosed Charge

$$Q_{enc} = \sigma A$$

Step 5: Apply Gauss's Law

$$2EA = \frac{\sigma A}{\epsilon_0}$$

$$\boxed{E = \frac{\sigma}{2\epsilon_0}}$$

Important Result: The field is uniform and independent of distance from an infinite plane sheet!

For a conducting plate (charge on one side only):

$$E = \frac{\sigma}{\epsilon_0}$$

Application 3: Electric Field Due to a Uniformly Charged Spherical Shell

Problem: Find the electric field at points (a) outside, (b) on the surface, and (c) inside a uniformly charged thin spherical shell of radius R and total charge Q .

Solution:

Case (a): Outside the Shell ($r > R$)

Gaussian Surface: Sphere of radius r ($> R$) concentric with the shell

By symmetry, E is radial and constant over the Gaussian surface.

$$\oint \vec{E} \cdot d\vec{A} = E \times 4\pi r^2$$

Enclosed charge = Q

$$E \times 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2} = \frac{kQ}{r^2}$$

The shell behaves as if all charge were concentrated at its center!

Case (b): On the Surface ($r = R$)

$$E = \frac{Q}{4\pi\epsilon_0 R^2} = \frac{\sigma}{\epsilon_0}$$

Where $\sigma = \frac{Q}{4\pi R^2}$ is the surface charge density.

Case (c): Inside the Shell ($r < R$)

Gaussian Surface: Sphere of radius r ($< R$)

Enclosed charge = 0 (all charge is on the shell's surface)

$$E \times 4\pi r^2 = \frac{0}{\epsilon_0}$$

$$E = 0$$

Remarkable Result: The electric field inside a uniformly charged spherical shell is zero!

Application 4: Electric Field Due to a Uniformly Charged Solid Sphere

Problem: Find the electric field at points (a) outside and (b) inside a uniformly charged solid sphere of radius R , total charge Q , and volume charge density ρ .

Solution:

Case (a): Outside the Sphere ($r > R$)

Same as spherical shell:

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

Case (b): Inside the Sphere ($r < R$)

Gaussian Surface: Sphere of radius r ($< R$)

Enclosed charge:

$$Q_{enc} = \rho \times \frac{4}{3}\pi r^3 = \frac{Q}{\frac{4}{3}\pi R^3} \times \frac{4}{3}\pi r^3 = Q \frac{r^3}{R^3}$$

Applying Gauss's law:

$$E \times 4\pi r^2 = \frac{Qr^3}{\varepsilon_0 R^3}$$

$$E = \frac{Qr}{4\pi\varepsilon_0 R^3}$$

Inside the sphere, E is proportional to r (increases linearly from center to surface).

1.7 Electric Potential - Complete Theory

Definition of Electric Potential

The electric potential at a point is the work done per unit positive charge by an external agent in bringing a test charge from infinity to that point against the electric field, without any acceleration. ^[1]

$$V = \frac{W_{\infty \rightarrow P}}{q_0}$$

Units: Volt (V) = Joule/Coulomb

Physical Interpretation: Electric potential represents the potential energy per unit charge. A positive charge naturally moves from higher to lower potential (like a ball rolling downhill).

Potential Difference

The potential difference between two points A and B is the work done per unit charge in moving a test charge from A to B:

$$V_B - V_A = \frac{W_{A \rightarrow B}}{q_0} = - \int_A^B \vec{E} \cdot d\vec{l}$$

1.8 Electric Potential Due to a Point Charge - Complete Derivation

Derivation

Consider a point charge Q at the origin. Find the potential at point P at distance r.

Step 1: The electric field at distance x from Q:

$$E = \frac{kQ}{x^2}$$

Step 2: Work done in moving a test charge q_0 from infinity to point P:

Work done against the electric force from x to $(x - dx)$:

$$dW = -\vec{F} \cdot d\vec{x} = -F dx \cos(180^\circ) = F dx = \frac{kQq_0}{x^2} dx$$

Note: The negative sign in $F \cdot dx$ becomes positive because the displacement is opposite to the force direction when bringing the charge from infinity.

Step 3: Total work from infinity to r :

$$W = \int_{\infty}^r \frac{kQq_0}{x^2} dx = kQq_0 \int_{\infty}^r x^{-2} dx$$
$$W = kQq_0 \left[-\frac{1}{x} \right]_{\infty}^r = kQq_0 \left(-\frac{1}{r} + \frac{1}{\infty} \right) = kQq_0 \left(-\frac{1}{r} + 0 \right)$$

Wait, let me recalculate properly:

$$W = - \int_{\infty}^r \vec{F} \cdot d\vec{r} = - \int_{\infty}^r \frac{kQq_0}{x^2} (-dx) = \int_{\infty}^r \frac{kQq_0}{x^2} dx$$
$$W = kQq_0 \left[-\frac{1}{x} \right]_{\infty}^r = kQq_0 \left(-\frac{1}{r} - 0 \right)$$

Actually, the correct approach:

$$W = \int_{\infty}^r F \cdot dr' = \int_{\infty}^r \frac{kQq_0}{r'^2} dr'$$

Since we're moving inward (from ∞ to r), and the force is outward for positive Q :

$$W_{external} = - \int_{\infty}^r \frac{kQq_0}{r'^2} dr' = kQq_0 \left[\frac{1}{r'} \right]_{\infty}^r = kQq_0 \left(\frac{1}{r} - 0 \right) = \frac{kQq_0}{r}$$

Step 4: Electric potential:

$$\boxed{V = \frac{W}{q_0} = \frac{kQ}{r} = \frac{Q}{4\pi\epsilon_0 r}}$$

Properties of Point Charge Potential

1. V is positive for positive Q , negative for negative Q
2. V decreases as distance increases (for positive Q)
3. $V = 0$ at infinity (reference point)
4. V is a scalar quantity (no direction)
5. Potential is spherically symmetric around a point charge

1.9 Potential Due to a System of Point Charges

Superposition Principle for Potential

The potential at any point due to a system of point charges is the algebraic (scalar) sum of potentials due to individual charges.

$$V = V_1 + V_2 + V_3 + \dots = \sum_{i=1}^n V_i = \sum_{i=1}^n \frac{kQ_i}{r_i}$$

$$V = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{Q_i}{r_i}$$

Where r_i is the distance from charge Q_i to the point where potential is calculated.

Advantage over Electric Field:

Potential is a scalar, so we simply add numbers (with signs). Electric field is a vector, requiring vector addition (more complex).

Example Problem

Two charges $Q_1 = +4 \mu\text{C}$ and $Q_2 = -2 \mu\text{C}$ are separated by 10 cm. Find the potential at the midpoint.

Solution:

Distance from each charge to midpoint: $r = 5 \text{ cm} = 0.05 \text{ m}$

$$V_1 = \frac{kQ_1}{r} = \frac{(9 \times 10^9)(4 \times 10^{-6})}{0.05} = 7.2 \times 10^5 \text{ V}$$

$$V_2 = \frac{kQ_2}{r} = \frac{(9 \times 10^9)(-2 \times 10^{-6})}{0.05} = -3.6 \times 10^5 \text{ V}$$

$$V_{total} = V_1 + V_2 = 7.2 \times 10^5 - 3.6 \times 10^5 = 3.6 \times 10^5 \text{ V}$$

1.10 Equipotential Surfaces - Complete Theory

Definition

An equipotential surface is a surface on which the electric potential has the same value at every point.

Properties of Equipotential Surfaces

1. No work is done in moving a charge along an equipotential surface:

$$W = q(V_B - V_A) = q \times 0 = 0$$

2. Electric field is always perpendicular to equipotential surfaces:

Proof: If the field had a component along the surface, work would be done in moving a charge along the surface, which contradicts property 1.

3. Two equipotential surfaces never intersect:

Proof: If they intersected, a point on the intersection would have two different potentials, which is impossible.

4. Equipotential surfaces are closer together where the field is stronger:

Since $E = -\frac{dV}{dr}$, larger E means V changes more rapidly with distance.

Examples of Equipotential Surfaces

Charge Distribution	Shape of Equipotential Surfaces
Point charge	Concentric spheres centered on charge
Infinite line charge	Coaxial cylinders around the line
Infinite plane	Parallel planes
Uniform field	Parallel planes perpendicular to field
Electric dipole	Complex curved surfaces

1.11 Relation Between Electric Field and Electric Potential

Electric Field as Negative Gradient of Potential

The electric field at any point is equal to the negative rate of change of potential with distance in that direction.

In one dimension:

$$E = -\frac{dV}{dr}$$

In three dimensions (Gradient form):

$$\vec{E} = -\nabla V = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$$

Derivation

Consider two points A and B separated by a small distance dr along the direction of the electric field.

Work done by electric field in moving charge q from A to B:

$$W = \vec{F} \cdot d\vec{r} = qE \cdot dr$$

Also, work done equals decrease in potential energy:

$$W = -q(V_B - V_A) = -q \cdot dV$$

Equating:

$$qE \cdot dr = -q \cdot dV$$

$$E = -\frac{dV}{dr}$$

Physical Interpretation

1. The negative sign indicates that E points from higher to lower potential
2. Electric field points in the direction of maximum decrease of potential
3. The magnitude of E equals the rate of change of V with distance
4. If V is constant, $E = 0$ (inside a conductor at equilibrium)

Component Form

The component of electric field in any direction equals the negative rate of change of potential in that direction:

$$E_x = -\frac{\partial V}{\partial x}, \quad E_y = -\frac{\partial V}{\partial y}, \quad E_z = -\frac{\partial V}{\partial z}$$

CHAPTER 2: CURRENT AND ELECTRICITY

2.1 Electric Current - Complete Theory

Definition

Electric current is the rate of flow of electric charge through a cross-section of a conductor.

$$I = \frac{dQ}{dt}$$

For steady current: $I = \frac{Q}{t}$

Units: Ampere (A) = Coulomb/second

Conventional Current Direction: The direction in which positive charges would flow (opposite to electron flow in metallic conductors).

Mechanism of Current Flow in Metals

1. Free electrons move randomly with high thermal velocities ($\sim 10^6$ m/s)
2. Without electric field: net charge flow = 0 (random motion)
3. With electric field: electrons acquire a small **drift velocity** superimposed on random motion
4. This drift produces a net charge flow = electric current

Drift Velocity

Definition: The average velocity acquired by free electrons in a conductor under the influence of an electric field.

$$v_d = \frac{eE\tau}{m}$$

Where:

- e = electron charge
- E = applied electric field
- τ = relaxation time (average time between collisions)
- m = electron mass

Typical value: $\sim 10^{-4}$ m/s (very small compared to thermal velocity)

Relation Between Current and Drift Velocity

Consider a conductor of cross-sectional area A with n free electrons per unit volume, each with charge e , moving with drift velocity v_d .

In time dt , electrons move a distance $v_d \cdot dt$.

Volume of electrons crossing a cross-section = $A \cdot v_d \cdot dt$

Number of electrons = $n \cdot A \cdot v_d \cdot dt$

Charge crossing = $n \cdot A \cdot v_d \cdot dt \cdot e$

Current:

$$I = neAv_d$$

2.2 Current Density

Definition

Current density is the current flowing per unit cross-sectional area perpendicular to the direction of current flow.

$$J = \frac{I}{A}$$

Units: A/m² (Ampere per square meter)

Vector Form

$$\vec{J} = nq\vec{v}_d$$

For electrons:

$$\vec{J} = -ne\vec{v}_d$$

The current through any surface S:

$$I = \int \vec{J} \cdot d\vec{A}$$

Relationship with Electric Field

$$\vec{J} = \sigma \vec{E}$$

Where σ is the conductivity of the material.

2.3 Resistance and Resistivity - Complete Theory

Resistance

Definition: Resistance is the opposition offered by a conductor to the flow of electric current.

$$R = \frac{V}{I}$$

Units: Ohm (Ω) = Volt/Ampere

Resistivity (Specific Resistance)

Definition: Resistivity is the resistance of a material per unit length and per unit cross-sectional area.

Relation between Resistance and Resistivity:

$$R = \rho \frac{L}{A}$$

Where:

- R = resistance (Ω)
- ρ = resistivity ($\Omega \cdot m$)
- L = length of conductor
- A = cross-sectional area

Physical Significance: Resistivity is a material property, independent of dimensions. It represents how strongly a material opposes current flow.

Conductivity

Definition: Conductivity is the reciprocal of resistivity.

$$\sigma = \frac{1}{\rho}$$

Units: Siemens per meter (S/m) or $(\Omega \cdot m)^{-1}$

Temperature Dependence of Resistance

For metals (conductors):

$$\rho_T = \rho_0 [1 + \alpha(T - T_0)]$$

$$R_T = R_0 [1 + \alpha(T - T_0)]$$

Where:

- α = temperature coefficient of resistance (positive for metals)
- T_0 = reference temperature
- R_0, ρ_0 = resistance and resistivity at reference temperature

For metals: $\alpha > 0$ (resistance increases with temperature)

For semiconductors: $\alpha < 0$ (resistance decreases with temperature)

2.4 Ohm's Law - Complete Theory

Statement

At constant temperature, the electric current flowing through a conductor is directly proportional to the potential difference across its ends.

$$V \propto I$$

$$\boxed{V = IR}$$

Microscopic Form of Ohm's Law

$$\boxed{\vec{J} = \sigma \vec{E}}$$

Or equivalently:

$$\vec{E} = \rho \vec{J}$$

Derivation of Ohm's Law from Drift Velocity

Step 1: Electric field in conductor: $E = \frac{V}{L}$

Step 2: Drift velocity: $v_d = \frac{eE\tau}{m} = \frac{eV\tau}{mL}$

Step 3: Current: $I = neAv_d = \frac{ne^2A\tau V}{mL}$

Step 4: Rearranging:

$$V = \frac{mL}{ne^2A\tau} \times I = RI$$

Where:

$$R = \frac{mL}{ne^2A\tau} = \rho \frac{L}{A}$$

And:

$$\rho = \frac{m}{ne^2\tau}$$

Ohmic and Non-Ohmic Conductors

Ohmic conductors: Follow Ohm's law (V-I graph is straight line through origin). Examples: Metallic conductors at constant temperature.

Non-Ohmic conductors: Do not follow Ohm's law. Examples: Diodes, transistors, electrolytes, discharge tubes.

2.5 Power in Electric Circuits

Definition

Electrical power is the rate at which electrical energy is converted to other forms (heat, light, mechanical work, etc.).

Derivation

Work done in moving charge Q through potential difference V :

$$W = QV$$

Power = Rate of doing work:

$$P = \frac{dW}{dt} = \frac{d(QV)}{dt} = V \frac{dQ}{dt} = VI$$

$$P = VI = I^2 R = \frac{V^2}{R}$$

Units: Watt (W) = Joule/second = Volt $\times$ Ampere

Electrical Energy

Energy consumed in time t :

$$E = Pt = VIt = I^2 Rt = \frac{V^2 t}{R}$$

Units: Joule (J) or kilowatt-hour (kWh)

Conversion: 1 kWh = 3.6×10^6 J

Joule's Law of Heating

When current flows through a resistance, electrical energy is converted to heat:

$$H = I^2 Rt$$

This is used in electric heaters, toasters, electric irons, etc.

2.6 Magnetic Force on Charged Particles and Current-Carrying Wires

Magnetic Force on a Moving Charge

When a charged particle moves through a magnetic field, it experiences a force called the Lorentz force.

$$\vec{F} = q(\vec{v} \times \vec{B})$$

Magnitude:

$$F = qvB \sin \theta$$

Where θ is the angle between $\vec{v}$ and $\vec{B}$.

Direction: Given by the right-hand rule (for positive charges)

- Point fingers in direction of $\vec{v}$
- Curl fingers toward $\vec{B}$
- Thumb points in direction of $\vec{F}$

For negative charges, force is opposite to thumb direction.

Special Cases:

- $\theta = 0^\circ$ or 180° : $F = 0$ (particle moves parallel to field)
- $\theta = 90^\circ$: $F = qvB$ (maximum force, circular motion)

Motion of Charged Particle in Magnetic Field

Case 1: $\mathbf{v} \perp \mathbf{B}$ (perpendicular)

The particle moves in a circle of radius:

$$r = \frac{mv}{qB}$$

Time period:

$$T = \frac{2\pi m}{qB}$$

Frequency (cyclotron frequency):

$$f = \frac{qB}{2\pi m}$$

Case 2: $\mathbf{v}$ at angle θ to $\mathbf{B}$

The particle moves in a helix (spiral path).

Magnetic Force on a Current-Carrying Wire

A current-carrying conductor in a magnetic field experiences a force because the moving charges (current) experience the Lorentz force.

$$\vec{F} = I(\vec{L} \times \vec{B})$$

Magnitude:

$$F = BIL \sin \theta$$

Where:

- I = current

- L = length of wire in the field
- B = magnetic field strength
- θ = angle between current direction and B

Direction: Right-hand rule (fingers from I toward B , thumb gives F)

2.7 Magnetic Field Lines

Definition

Magnetic field lines are imaginary curves used to represent the magnetic field. The tangent at any point gives the direction of the magnetic field at that point.

Properties of Magnetic Field Lines

1. **Form closed loops:** Unlike electric field lines, magnetic field lines always form closed loops (no magnetic monopoles exist)
2. **Never intersect:** If they did, the field would have two directions at one point
3. **Direction:** Outside a magnet: North to South. Inside a magnet: South to North
4. **Density proportional to field strength:** More lines per unit area = stronger field
5. **Lines never start or end:** They form continuous loops

Magnetic Field Due to Common Configurations

Configuration	Field Pattern
Straight current-carrying wire	Concentric circles around wire
Circular loop	Similar to bar magnet
Solenoid	Uniform inside, like bar magnet outside
Bar magnet	Emerge from N-pole, enter S-pole

2.8 Biot-Savart Law - Complete Derivation and Applications

Statement

The Biot-Savart law gives the magnetic field produced by a small current element. The magnetic field dB at a point P due to a current element Idl is: ^[1]

1. Directly proportional to the current I
2. Directly proportional to the length element dl
3. Directly proportional to $\sin \theta$ (angle between dl and r)
4. Inversely proportional to the square of distance r

Mathematical Expression

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

Magnitude form:

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}$$

Where:

- $\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$ (permeability of free space)
- I = current (A)
- dl = length element (m)
- r = distance from element to point P (m)
- θ = angle between dl and the line joining dl to P

Direction of dB

The direction of dB is given by the right-hand rule:

- Point fingers along current direction (dl)
- Curl fingers toward position vector (r)
- Thumb points in direction of dB

Alternatively: dB is perpendicular to the plane containing dl and r .

Application: Magnetic Field Due to a Long Straight Current-Carrying Wire

Problem: Find the magnetic field at perpendicular distance r from an infinitely long straight wire carrying current I .

Solution:

Step 1: Setup

Consider a small element dl at distance l from the foot of perpendicular from point P to the wire.

Distance from element to P: $R = \sqrt{r^2 + l^2}$

Angle: $\sin \theta = \frac{r}{R} = \frac{r}{\sqrt{r^2 + l^2}}$

Step 2: Apply Biot-Savart Law

$$dB = \frac{\mu_0 I}{4\pi} \frac{dl \sin \theta}{R^2} = \frac{\mu_0 I}{4\pi} \frac{dl \cdot r}{(r^2 + l^2)^{3/2}}$$

Step 3: Integrate over entire wire

$$B = \frac{\mu_0 I r}{4\pi} \int_{-\infty}^{+\infty} \frac{dl}{(r^2 + l^2)^{3/2}}$$

Using substitution $l = r \tan \phi$:

$$B = \frac{\mu_0 I r}{4\pi} \times \frac{2}{r^2} = \frac{\mu_0 I}{2\pi r}$$

$$\boxed{B = \frac{\mu_0 I}{2\pi r}}$$

Direction: Circular, around the wire (right-hand thumb rule)

Application: Magnetic Field at the Center of a Circular Loop

For a circular loop of radius R carrying current I , the magnetic field at the center:

$$\boxed{B = \frac{\mu_0 I}{2R}}$$

For N turns:

$$\boxed{B = \frac{\mu_0 N I}{2R}}$$

2.9 Two Parallel Current-Carrying Conductors

Force Between Two Parallel Wires

Consider two long parallel wires separated by distance d , carrying currents I_1 and I_2 .

Step 1: Wire 1 creates a magnetic field at wire 2:

$$B_1 = \frac{\mu_0 I_1}{2\pi d}$$

Step 2: Force on wire 2 (length L) due to B_1 :

$$F = B_1 I_2 L = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$

Force per unit length:

$$\boxed{\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}}$$

Direction of Force

- **Same direction currents:** Attract each other
- **Opposite direction currents:** Repel each other

Physical Explanation: Consider the magnetic field created by one wire at the location of the other. Using the right-hand rule for force on a current-carrying wire, you can verify the attraction/repulsion.

Definition of Ampere

One ampere is that steady current which, when maintained in two straight parallel conductors of infinite length and negligible cross-section, placed one meter apart in vacuum, produces a force of 2×10^{-7} N per meter length between them.

2.10 Ampere's Circuital Law - Complete Theory

Statement

The line integral of magnetic field around any closed loop (Amperian loop) is equal to μ_0 times the total current enclosed by the loop.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

Application 1: Magnetic Field Inside a Solenoid

Solenoid: A long coil of wire wound in the form of a helix.

Assumptions:

1. The solenoid is long (length $\gg$ radius)
2. Field inside is uniform and parallel to axis
3. Field outside is negligible

Derivation:

Consider a rectangular Amperian loop ABCD:

- AB: Inside the solenoid, parallel to axis (length = L)
- CD: Outside the solenoid ($B \approx 0$)
- BC, DA: Perpendicular to field

$$\begin{aligned}\oint \vec{B} \cdot d\vec{l} &= \int_{AB} B dl + \int_{BC} + \int_{CD} + \int_{DA} \\ &= BL + 0 + 0 + 0 = BL\end{aligned}$$

Current enclosed = $nL \times I$ (where n = turns per unit length)

Applying Ampere's law:

$$BL = \mu_0 nLI$$

$$\boxed{B = \mu_0 nI}$$

For N total turns over length L:

$$\boxed{B = \frac{\mu_0 NI}{L}}$$

Application 2: Magnetic Field of a Toroid

Toroid: A solenoid bent into a circular ring shape.

Consider a circular Amperian loop of radius r inside the toroid:

$$\oint \vec{B} \cdot d\vec{l} = B \times 2\pi r$$

Current enclosed = NI (N = total turns)

$$B \times 2\pi r = \mu_0 NI$$

$$\boxed{B = \frac{\mu_0 NI}{2\pi r}}$$

Note: Field is zero outside the toroid and inside the inner radius.

2.11 Faraday's Experiments and Laws of Electromagnetic Induction

Faraday's Experiments^[1]

Michael Faraday conducted three groundbreaking experiments in 1831:

Experiment 1: Moving Magnet

- A magnet was moved toward a coil connected to a galvanometer
- When magnet moved toward coil: galvanometer deflected one way
- When magnet moved away: galvanometer deflected opposite way
- When magnet was stationary: no deflection

Conclusion: A changing magnetic flux through a coil induces an EMF.

Experiment 2: Electromagnetic Induction

- Two coils were placed close together
- Primary coil connected to battery through switch
- Secondary coil connected to galvanometer

- When switch was closed (current started): galvanometer deflected momentarily
- When switch was opened (current stopped): galvanometer deflected in opposite direction

Conclusion: A changing current in one coil induces an EMF in a nearby coil.

Experiment 3: Relative Motion

- Same setup as Experiment 1
- When coil moved toward stationary magnet: current induced
- When either coil or magnet moves: current is induced

Conclusion: It is the relative motion between the coil and magnet that matters, not which one is moving.

Faraday's First Law of Electromagnetic Induction^[1]

Statement: Whenever the magnetic flux linked with a circuit changes, an electromotive force (EMF) is induced in the circuit. The induced EMF lasts as long as the change in magnetic flux continues.

Ways to change magnetic flux:

1. Moving a magnet relative to a coil
2. Moving a coil in a magnetic field
3. Changing the area of a coil in a magnetic field
4. Rotating a coil in a magnetic field
5. Changing the magnetic field strength

Faraday's Second Law of Electromagnetic Induction^[1]

Statement: The magnitude of the induced EMF is equal to the rate of change of magnetic flux linkage with the circuit.

$$\varepsilon = -N \frac{d\Phi_B}{dt}$$

Where:

- ε = induced EMF (Volts)
- N = number of turns in the coil
- Φ_B = magnetic flux through each turn (Weber)
- $N\Phi_B$ = total flux linkage

Complete Derivation of Faraday's Law^[1]

Step 1: At time t_1 , flux linkage with coil = $N\Phi_1$

Step 2: At time t_2 , flux linkage with coil = $N\Phi_2$

Step 3: Change in flux linkage = $N(\Phi_2 - \Phi_1) = N\Delta\Phi$

Step 4: Rate of change of flux linkage = $N\Delta\Phi/\Delta t$

Step 5: Taking the derivative (instantaneous rate):

$$\varepsilon = -N \frac{d\Phi}{dt}$$

The negative sign indicates that the induced EMF opposes the change (Lenz's Law).

Methods to Change Magnetic Flux

Since $\Phi = B \cdot A \cdot \cos \theta$, flux can be changed by changing:

1. **Magnetic field strength (B):** Moving magnet closer/farther
2. **Area (A):** Stretching or compressing the loop
3. **Angle (θ):** Rotating the loop in the field
4. **Any combination of above**

2.12 Lenz's Law - Complete Explanation

Statement

The direction of the induced EMF (and hence the induced current) is always such that it opposes the change in magnetic flux that produces it.

Mathematical Representation

The negative sign in Faraday's law represents Lenz's law:

$$\varepsilon = -N \frac{d\Phi}{dt}$$

Physical Explanation

Case 1: Increasing Flux (North pole approaching coil)

- Flux through coil increases
- Induced current creates its own magnetic field
- This field opposes the increase (points opposite to approaching pole)
- Coil acts like a magnet with North pole facing the approaching magnet

- This creates repulsion, opposing the approach

Case 2: Decreasing Flux (North pole moving away)

- Flux through coil decreases
- Induced current opposes this decrease
- Induced field is in same direction as original field
- Coil acts like a magnet with South pole facing the receding North pole
- This creates attraction, opposing the separation

Energy Conservation

Lenz's law is a consequence of conservation of energy:

- Work must be done against the opposing force to change the flux
- This work appears as electrical energy in the induced current
- If the induced current aided the change, energy would be created from nothing (violating conservation)

Applications of Lenz's Law

1. **Eddy current braking:** Opposing currents slow down moving conductors
2. **Induction cookers:** Energy is transferred to the pan
3. **Metal detectors:** Change in field detected by opposing currents
4. **Electromagnetic damping:** Used in galvanometers

2.13 Motional EMF - Complete Derivation

Definition

When a conductor moves through a magnetic field, an EMF is induced across its ends. This is called motional EMF.

Derivation for a Rod Moving in a Magnetic Field

Consider a conducting rod of length L moving with velocity v perpendicular to a uniform magnetic field B .

Method 1: Using Lorentz Force

Step 1: Each free electron in the rod experiences a magnetic force:

$$\vec{F} = q(\vec{v} \times \vec{B})$$

For an electron: $F = evB$ (perpendicular to both v and B)

Step 2: This force pushes electrons toward one end of the rod, creating a charge separation.

Step 3: The charge separation creates an electric field E inside the rod.

Step 4: At equilibrium, electric force = magnetic force:

$$eE = evB$$

$$E = vB$$

Step 5: EMF = Electric field $\times$ Length:

$$\boxed{\varepsilon = EL = BLv}$$

Method 2: Using Faraday's Law

Consider the rod as part of a rectangular loop with the rod as one side.

Step 1: In time dt , the rod moves distance $dx = v \cdot dt$

Step 2: Change in area of loop: $dA = L \cdot dx = Lv \cdot dt$

Step 3: Change in flux: $d\Phi = B \cdot dA = BLv \cdot dt$

Step 4: Induced EMF:

$$\varepsilon = -\frac{d\Phi}{dt} = -\frac{BLv \cdot dt}{dt} = BLv$$

$$\boxed{\varepsilon = BLv}$$

General Case: Rod Moving at Angle θ to Field

$$\varepsilon = BLv \sin \theta$$

Where θ is the angle between v and B .

2.14 Induced Electric Field

Concept

Faraday's law reveals a profound truth: a changing magnetic field creates an electric field, even in the absence of charges!

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

Properties of Induced Electric Field

Property	Electrostatic Field	Induced Electric Field
Source	Electric charges	Changing magnetic field
Field lines	Start/end on charges	Form closed loops
Conservative?	Yes ($\oint \mathbf{E} \cdot d\mathbf{l} = 0$)	No ($\oint \mathbf{E} \cdot d\mathbf{l} \neq 0$)
Potential	Can be defined	Cannot be defined uniquely

Significance

The induced electric field is fundamentally different from the electrostatic field:

1. It exists even in empty space (no charges needed)
2. Its field lines form closed loops (no beginning or end)
3. It is non-conservative (work done around closed path $\neq 0$)

2.15 Displacement Current and Maxwell's Equations

The Problem with Ampere's Law^[1]

Consider a capacitor being charged by a current I :

- Current flows in the wires connected to the capacitor
- But NO current flows between the capacitor plates (it's an insulator!)

Apply Ampere's law to two surfaces with the same boundary:

1. **Surface through wire:** Enclosed current = I
2. **Surface through capacitor gap:** Enclosed current = 0

This gives contradictory results! Ampere's law seems to fail.

Maxwell's Solution: Displacement Current^[1]

Maxwell proposed that a changing electric field between the capacitor plates is equivalent to a current. He called this the **displacement current**.

Derivation of Displacement Current^[1]

Step 1: For a charging capacitor, the charge on plates at any instant:

$$q = \epsilon_0 EA$$

Where E is the electric field between plates and A is the plate area.

Step 2: The conduction current in the wire:

$$I_c = \frac{dq}{dt} = \varepsilon_0 A \frac{dE}{dt} = \varepsilon_0 \frac{d(EA)}{dt} = \varepsilon_0 \frac{d\Phi_E}{dt}$$

Step 3: Maxwell defined displacement current:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

Physical Meaning: A changing electric flux through a surface is equivalent to a current through that surface, as far as producing magnetic field is concerned.

Modified Ampere's Law (Ampere-Maxwell Law)

$$\oint \vec{B} \cdot d\vec{l} = \mu_0(I_c + I_d) = \mu_0 I_c + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

This is valid for all situations, including capacitors.

Induced Magnetic Field

Just as a changing magnetic field induces an electric field (Faraday's law), a changing electric field induces a magnetic field (Maxwell's addition to Ampere's law).

This symmetry between electric and magnetic fields led Maxwell to predict electromagnetic waves!

Maxwell's Equations - The Complete Set

These four equations describe all of electromagnetic phenomena:

1. Gauss's Law for Electricity:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\varepsilon_0}$$

Electric charges produce electric fields

2. Gauss's Law for Magnetism:

$$\oint \vec{B} \cdot d\vec{A} = 0$$

No magnetic monopoles exist; magnetic field lines are closed loops

3. Faraday's Law:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

A changing magnetic field produces an electric field

4. Ampere-Maxwell Law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

Electric currents and changing electric fields produce magnetic fields

Significance of Maxwell's Equations

1. They unify electricity and magnetism into electromagnetism
2. They predict the existence of electromagnetic waves
3. They show that light is an electromagnetic wave
4. They form the foundation of all electrical and electronic technology

CHAPTER 3: OPTICS

3.1 Refraction of Light Waves - Complete Theory

Definition

Refraction is the bending of light as it passes from one transparent medium to another with different optical densities.

Cause of Refraction

Light travels at different speeds in different media. When a wavefront enters a new medium at an angle, one part of the wavefront slows down (or speeds up) before the other, causing the wavefront to bend.

Laws of Refraction (Snell's Law)

First Law: The incident ray, refracted ray, and the normal to the interface at the point of incidence all lie in the same plane.

Second Law (Snell's Law):

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Or:

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1} = \frac{v_1}{v_2}$$

Where:

- n_1, n_2 = refractive indices of media 1 and 2
- θ_1 = angle of incidence

- θ_2 = angle of refraction
- v_1, v_2 = speeds of light in media 1 and 2

Refractive Index

Absolute Refractive Index:

$$n = \frac{c}{v} = \frac{\text{speed of light in vacuum}}{\text{speed of light in medium}}$$

Relative Refractive Index:

$$n_{21} = \frac{n_2}{n_1} = \frac{v_1}{v_2}$$

Important Results

1. When light goes from rarer to denser medium ($n_1 < n_2$): It bends toward the normal ($\theta_2 < \theta_1$)
2. When light goes from denser to rarer medium ($n_1 > n_2$): It bends away from normal ($\theta_2 > \theta_1$)
3. If $\theta_1 = 0$ (normal incidence): No bending occurs

Refractive Indices of Common Materials

Material	Refractive Index
Vacuum	1.000 (exact)
Air	1.0003
Water	1.33
Glass	1.5
Diamond	2.42

3.2 Total Internal Reflection - Complete Theory

Definition

Total internal reflection (TIR) is the complete reflection of light at the boundary between two media when light travels from a denser to a rarer medium and the angle of incidence exceeds the critical angle.

Conditions for Total Internal Reflection

Condition 1: Light must travel from a denser medium to a rarer medium ($n_1 > n_2$)

Condition 2: The angle of incidence must be greater than the critical angle ($\theta > \theta_c$)

Critical Angle

Definition: The critical angle is the angle of incidence in the denser medium for which the angle of refraction in the rarer medium is 90° .

Derivation:

At critical angle, $\theta_2 = 90^\circ$

Using Snell's law:

$$n_1 \sin \theta_c = n_2 \sin 90^\circ$$

$$n_1 \sin \theta_c = n_2$$

$$\sin \theta_c = \frac{n_2}{n_1}$$

For light going from medium to air ($n_2 = 1$):

$$\sin \theta_c = \frac{1}{n}$$

Critical Angles for Different Materials

Material	Critical Angle (with air)
Water ($n=1.33$)	48.6°
Glass ($n=1.5$)	41.8°
Diamond ($n=2.42$)	24.4°

What Happens at Different Angles?

Angle of Incidence	Result
$\theta < \theta_c$	Both refraction and partial reflection
$\theta = \theta_c$	Refracted ray grazes the surface ($\theta_2 = 90^\circ$)
$\theta > \theta_c$	Total internal reflection (no refraction)

Applications of Total Internal Reflection

1. **Optical Fibers:** Light travels through glass fiber by repeated TIR, enabling high-speed data transmission
2. **Prisms:** Right-angle prisms use TIR to reflect light with 100% efficiency (better than mirrors)
3. **Diamonds:** Low critical angle causes multiple internal reflections, creating sparkle
4. **Endoscopes:** Medical instruments to view inside the body
5. **Mirages:** Caused by TIR in layers of air with different temperatures

3.3 Interference of Light - Complete Theory

Definition

Interference is the phenomenon in which two or more light waves superpose to form a resultant wave with a redistribution of intensity.

Principle of Superposition

When two or more waves meet at a point, the resultant displacement is the vector sum of individual displacements.

$$\vec{y}_{resultant} = \vec{y}_1 + \vec{y}_2 + \dots$$

Coherent Sources

Definition: Two sources are coherent if they emit light of the same frequency and maintain a constant phase relationship.

Requirements for sustained interference:

1. Same frequency (or wavelength)
2. Constant phase difference
3. Same or nearly same amplitude (for good contrast)
4. Same state of polarization

Note: Two independent light sources (like two bulbs) are never coherent because light emission is random.

Types of Interference

1. Constructive Interference:

- Occurs when waves are in phase (crest meets crest)
- Phase difference: $0, 2\pi, 4\pi, \dots$ (or $0, \lambda, 2\lambda, \dots$)
- Path difference: $0, \lambda, 2\lambda, 3\lambda, \dots = n\lambda$
- Resultant amplitude: $A_1 + A_2$ (maximum)
- Resultant intensity: Maximum (bright fringe)

2. Destructive Interference:

- Occurs when waves are out of phase (crest meets trough)
- Phase difference: $\pi, 3\pi, 5\pi, \dots$ (or $\lambda/2, 3\lambda/2, \dots$)
- Path difference: $\lambda/2, 3\lambda/2, 5\lambda/2, \dots = (n + \frac{1}{2})\lambda$
- Resultant amplitude: $|A_1 - A_2|$ (minimum)

- Resultant intensity: Minimum (dark fringe)

Mathematical Analysis

For two waves of same amplitude A:

$$y_1 = A \sin(\omega t)$$

$$y_2 = A \sin(\omega t + \phi)$$

Resultant:

$$y = y_1 + y_2 = 2A \cos\left(\frac{\phi}{2}\right) \sin\left(\omega t + \frac{\phi}{2}\right)$$

Resultant amplitude:

$$A_R = 2A \cos\left(\frac{\phi}{2}\right)$$

Resultant intensity ($I \propto A^2$):

$$I = 4I_0 \cos^2\left(\frac{\phi}{2}\right)$$

Where I_0 is the intensity due to each source.

3.4 Young's Double-Slit Experiment - Complete Theory and Derivation

Historical Significance

In 1801, Thomas Young performed this experiment which provided the first definitive proof that light behaves as a wave. This experiment settled the debate between Newton's particle theory and Huygens' wave theory in favor of waves.^[2]

Experimental Setup

1. Monochromatic light source (single wavelength)
2. Single slit S_0 (to create a coherent source)
3. Double slit barrier with two narrow slits S_1 and S_2 separated by distance d
4. Screen at distance D from the double slit

Theory of Double-Slit Interference^[2]

Step 1: Coherent Sources

Light passing through slits S_1 and S_2 acts as two coherent sources (same frequency, constant phase relationship).

Step 2: Path Difference

Consider a point P on the screen at angle θ from the central axis.

The path difference between waves from S_1 and S_2 to point P:

$$\Delta = S_2P - S_1P = d \sin \theta$$

For small angles (when $D \gg d$): $\sin \theta \approx \tan \theta = \frac{y}{D}$

$$\Delta = \frac{dy}{D}$$

Where y is the distance of point P from the center of the screen.

Step 3: Conditions for Bright and Dark Fringes

Constructive Interference (Bright Fringes): ^[2]

Path difference = whole number of wavelengths

$$d \sin \theta = m\lambda$$

Where $m = 0, \pm 1, \pm 2, \pm 3, \dots$ (order of the fringe)

Destructive Interference (Dark Fringes): ^[2]

Path difference = odd multiple of half wavelength

$$d \sin \theta = \left(m + \frac{1}{2}\right)\lambda$$

Position of Fringes

Position of mth Bright Fringe:

$$y_m = \frac{m\lambda D}{d}$$

For $m = 0$: Central bright fringe ($y_0 = 0$)

For $m = 1$: First bright fringe, etc.

Position of mth Dark Fringe:

$$y_m = \frac{(2m + 1)\lambda D}{2d}$$

Fringe Width (β)

Definition: The distance between two consecutive bright (or dark) fringes.

Derivation:

Distance between mth and (m+1)th bright fringe:

$$\beta = y_{m+1} - y_m = \frac{(m+1)\lambda D}{d} - \frac{m\lambda D}{d}$$

$$\boxed{\beta = \frac{\lambda D}{d}}$$

Important Observations:

1. Fringe width is same for all fringes (uniform spacing)
2. $\beta \propto \lambda$ (red light gives wider fringes than blue)
3. $\beta \propto D$ (fringes spread out as screen moves away)
4. $\beta \propto 1/d$ (closer slits give wider fringes)

Intensity Distribution

The intensity at any point on the screen:

$$I = 4I_0 \cos^2 \left(\frac{\pi d \sin \theta}{\lambda} \right)$$

Or in terms of position y:

$$I = 4I_0 \cos^2 \left(\frac{\pi dy}{\lambda D} \right)$$

Where I_0 is the intensity due to each slit alone.

Solved Example^[2]

Problem: Light of wavelength 633 nm passes through double slits separated by 0.0100 mm. Find the angle for the third bright fringe.

Solution:

For third bright fringe, $m = 3$

Using $d \sin \theta = m\lambda$:

$$\sin \theta = \frac{m\lambda}{d} = \frac{3 \times 633 \times 10^{-9}}{0.0100 \times 10^{-3}} = 0.190$$

$$\theta = \sin^{-1}(0.190) = 10.95^\circ$$

3.5 Interference from Thin Films - Complete Theory

Phenomenon

When light falls on a thin transparent film (like a soap bubble or oil film on water), interference occurs between rays reflected from the top and bottom surfaces.

Path Difference Analysis

Consider a thin film of thickness t and refractive index μ .

Step 1: Ray 1 reflects from top surface

Step 2: Ray 2 enters the film, reflects from bottom surface, and emerges

Optical Path Difference:

$$\Delta = 2\mu t \cos r$$

Where r is the angle of refraction inside the film.

Phase Change on Reflection

Important: When light reflects from an optically denser medium, it undergoes a phase change of π (equivalent to path difference of $\lambda/2$).

- Reflection at air-film interface: Phase change of π (if film is denser)
- Reflection at film-air interface: No phase change

Conditions for Interference (Reflected Light)

Taking into account the phase change at the first surface:

Constructive Interference (Bright):

$$2\mu t \cos r = \left(m + \frac{1}{2}\right)\lambda$$

Destructive Interference (Dark):

$$2\mu t \cos r = m\lambda$$

Where $m = 0, 1, 2, 3, \dots$

Note: These conditions are reversed for transmitted light.

For Normal Incidence ($r = 0$)

Bright: $2\mu t = \left(m + \frac{1}{2}\right)\lambda$

Dark: $2\mu t = m\lambda$

Applications

1. **Anti-reflection coatings:** Thin film coating on lenses to reduce reflection
2. **Colors in soap bubbles:** Different thicknesses produce different colors
3. **Oil film colors:** Thin oil layer on water shows spectrum of colors
4. **Newton's rings:** Air film between lens and glass plate

3.6 Diffraction - Complete Theory

Definition

Diffraction is the bending of light waves around obstacles or the spreading of light through narrow openings.

Difference from Interference

Interference	Diffraction
Superposition of waves from two sources	Superposition of waves from different parts of same wavefront
Fringes have equal intensity	Central maximum is brightest, others fade
Fringes are equally spaced	Fringes are not equally spaced
Requires two coherent sources	Single slit or obstacle is sufficient

Types of Diffraction

1. Fresnel Diffraction:

- Source and/or screen at finite distance
- Wavefronts are spherical
- More complex analysis
- Example: Shadow of a razor blade

2. Fraunhofer Diffraction:

- Source and screen at infinite distance (or using lenses)
- Wavefronts are plane waves
- Simpler analysis
- Example: Single slit with collimated light

3.7 Single-Slit Diffraction - Complete Derivation

Setup

Monochromatic light of wavelength λ passes through a single slit of width a . The diffraction pattern is observed on a screen at distance D .

Analysis Using Huygens' Principle

Step 1: Divide the slit into many small elements, each acting as a source of secondary wavelets.

Step 2: These wavelets interfere with each other at points on the screen.

Step 3: At the central point ($\theta = 0$): All wavelets travel equal distances, constructive interference
→ bright central maximum.

Step 4: At other points: Path difference between wavelets from different parts of slit causes interference.

Condition for Diffraction Minima (Dark Fringes)

Method: Divide slit into pairs of elements

Consider light at angle θ . Divide the slit into two equal halves.

For each point in the upper half, there's a corresponding point in the lower half at distance $a/2$.

Path difference between these pairs: $\frac{a}{2} \sin \theta$

For destructive interference, this path difference = $\lambda/2$:

$$\frac{a}{2} \sin \theta = \frac{\lambda}{2}$$

$$a \sin \theta = \lambda$$

→ First minimum

General condition for minima:

$$a \sin \theta = m\lambda$$

Where $m = \pm 1, \pm 2, \pm 3, \dots$ (NOT zero, since $m = 0$ is central maximum)

Position of Minima on Screen

For small angles: $\sin \theta \approx \tan \theta = \frac{y}{D}$

Position of m th minimum:

$$y_m = \frac{m\lambda D}{a}$$

Condition for Secondary Maxima

Between the minima, there are secondary maxima (weaker bright fringes).

Approximate condition:

$$a \sin \theta = \left(m + \frac{1}{2}\right) \lambda$$

Where $m = 1, 2, 3, \dots$

Width of Central Maximum

The central maximum extends from first minimum on one side to first minimum on other side.

Angular width:

$$2\theta_1 = 2 \sin^{-1} \left(\frac{\lambda}{a} \right) \approx \frac{2\lambda}{a}$$

(for small angles)

Linear width on screen:

$$W = \frac{2\lambda D}{a}$$

Important: The central maximum is twice as wide as other maxima.

Intensity Distribution

The intensity at angle θ :

$$I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2$$

Where:

$$\alpha = \frac{\pi a \sin \theta}{\lambda}$$

At central maximum ($\theta = 0$): $I = I_0$ (maximum intensity)

At first secondary maximum: $I \approx 0.045 I_0$ (only 4.5% of central)

At second secondary maximum: $I \approx 0.016 I_0$ (only 1.6% of central)

Comparison: Single Slit vs Double Slit

Property	Single Slit	Double Slit
Central maximum	Very bright, wide	Bright, narrow
Other maxima	Much weaker	Nearly equal to central
Fringe width	Non-uniform	Uniform
Depends on	Slit width (a)	Slit separation (d)
Formula for dark	$a \sin \theta = m\lambda$	$d \sin \theta = (m + \frac{1}{2})\lambda$

3.8 Polarization of Light - Complete Theory

Nature of Light Waves

Light is a transverse electromagnetic wave consisting of oscillating electric and magnetic fields perpendicular to the direction of propagation.

Unpolarized Light: The electric field vector vibrates in all directions perpendicular to the direction of propagation (natural light from sun, bulb, etc.).

Polarized Light: The electric field vector vibrates in only one specific plane.

Definition of Polarization

Polarization is the phenomenon of restricting the vibrations of light (electric field vector) to a single plane.

Types of Polarization

1. Linear (Plane) Polarization:

Electric field oscillates in a single plane. The direction of oscillation is called the polarization direction.

2. Circular Polarization:

The tip of the electric field vector traces a circle as the wave propagates. Results from two perpendicular linear polarizations with 90° phase difference and equal amplitudes.

3. Elliptical Polarization:

The tip of the electric field vector traces an ellipse. Results from two perpendicular linear polarizations with arbitrary phase difference and/or unequal amplitudes.

Methods of Polarization

1. Polarization by Selective Absorption (Dichroism):

- Some crystals (tourmaline) or synthetic materials (Polaroid) absorb one component of vibration while transmitting the perpendicular component.

2. Polarization by Reflection:

- When unpolarized light reflects from a surface at Brewster's angle, the reflected light is completely polarized.
- **Brewster's Angle:** $\tan \theta_B = n$

3. Polarization by Refraction:

- The refracted ray is partially polarized.

4. Polarization by Double Refraction:

- Certain crystals (calcite, quartz) split incident light into two polarized rays (ordinary and extraordinary rays).

5. Polarization by Scattering:

- Light scattered by small particles (like atmospheric molecules) is partially polarized.
- This is why the sky appears polarized (useful for some animals' navigation).

3.9 Polarizing Sheets (Polaroids) and Malus's Law

Polaroid (Polarizing Sheet)

Definition: A Polaroid is a synthetic material that transmits light polarized in one direction while absorbing light polarized perpendicular to that direction.

Construction: Made by embedding long-chain molecules (like iodine-treated polyvinyl alcohol) in a plastic sheet. These chains are aligned parallel to each other.

Mechanism: The aligned molecules absorb the electric field component parallel to them while allowing the perpendicular component to pass.

Polarization Direction: The direction perpendicular to the aligned chains (the direction in which light is transmitted).

Terminology

Polarizer: The first Polaroid that converts unpolarized light to polarized light.

Analyzer: A second Polaroid used to analyze or verify polarization.

Transmission Axis: The direction along which polarized light is transmitted.

Effect of Polaroid on Unpolarized Light

When unpolarized light passes through a Polaroid:

- The transmitted light is linearly polarized
- Intensity is reduced to half: $I = \frac{I_0}{2}$

This is because, on average, half the intensity is in each perpendicular direction.

Malus's Law - Complete Derivation

Statement: When plane-polarized light passes through an analyzer, the intensity of transmitted light is proportional to the square of the cosine of the angle between the polarization direction and the analyzer's transmission axis.

Derivation:

Step 1: Let polarized light with electric field amplitude E_0 and intensity I_0 be incident on an analyzer at angle θ to its transmission axis.

Step 2: Resolve E_0 into components:

- Parallel to transmission axis: $E_0 \cos \theta$ (transmitted)
- Perpendicular to transmission axis: $E_0 \sin \theta$ (absorbed)

Step 3: Transmitted amplitude:

$$E = E_0 \cos \theta$$

Step 4: Since intensity $\propto$ amplitude²:

$$I = I_0 \cos^2 \theta$$

$$\boxed{I = I_0 \cos^2 \theta}$$

This is **Malus's Law**.

Special Cases of Malus's Law

Angle (θ)	$\cos^2 \theta$	Transmitted Intensity
0°	1	I_0 (maximum)
30°	0.75	$0.75 I_0$
45°	0.5	$0.5 I_0$
60°	0.25	$0.25 I_0$
90°	0	0 (no transmission)

Applications of Polaroids

1. **Sunglasses:** Reduce glare from reflected light (which is partially polarized)
2. **LCD displays:** Use polarizers to control light transmission
3. **Photography:** Polarizing filters reduce reflections and enhance contrast
4. **3D movies:** Different polarizations for left and right eye images
5. **Stress analysis:** Photoelastic stress analysis uses polarized light
6. **Optical instruments:** Reduce stray light

Solved Problems on Polarization

Problem 1: Unpolarized light of intensity I_0 passes through three polarizers. The first and third have their transmission axes vertical. The middle one has its axis at 45° to vertical. Find the final intensity.

Solution:

After first polarizer: $I_1 = \frac{I_0}{2}$ (polarized vertical)

After second polarizer (at 45°): $I_2 = I_1 \cos^2(45^\circ)$



1. CamScanner-13-09-2025-13.27_removed.pdf

2. <https://www2.tntech.edu/leap/murdock/books/v4chap3.pdf>